Example (1.1):

(a) The temperature of the skin is approximately 35.0°C. At what wavelength does the radiation emitted from the skin reach its peak?

(b) Find the wavelength corresponding to the peak of the radiation curve for the heating element of an electric oven at a temperature of $1.20 \times 10^3 K$.

Solution:

(a) Apply Wien's displacement law:

$$\lambda_{max} T = 2.898 \times 10^{-3}\ mk$$

Body absolute temperature = 273 + 35 =308 K

$$\lambda_{max} = \frac{2.898 \times 10^{-3}}{308} = 9.41 \times 10^{-6}\ m = 9.41\ \ \mu m$$

Example (1.2):

The peak in the radiation curve of the Sun is 510 nm. Calculate the temperature of the surface of the Sun.

Solution:

The maximum wavelength $\lambda_{max} = 510 \times 10^{-9}\ m$

To find temperature of the Sun surface apply Wien's displacement law:

$$T = \frac{2.898 \times 10^{-3}}{510 \times 10^{-9}} = 5682\ K\ \approx 5700\ K$$

Example (1.3):

Assume that stars behave like black body, for the sun $\lambda_{max} = 5100\ \ A^o$ whereas for the north-star $\lambda_{max} = 3500\ \ A^o$ find:

(a) The surface temperature of these stars.

(b) Using Stefan's Boltzmann law and the results obtained above; determine the power radiation from 1 cm^2 of star surface. Given $\sigma = 5.67 \times 10^{-8}\ Wm^{-2}K^{-4}$

Solution:

(a) Using Wien's displacement law to find surface temperature of these stars:

$$\lambda_{max} T = 2.898 \times 10^{-3}\ mk$$

- For the sun $T = \frac{2.898 \times 10^{-3}}{5100 \times 10^{-10}} = 5700\ K$

- For the north-star $T = \frac{2.898\times10^{-3}}{3500\times10^{-10}} = 8300\ K$

(b) Using Stefan's Boltzmann law to determine the power radiation from star surface:

$$I = \sigma T^4$$

- For the sun $I = (5.67 \times 10^{-8})(5700)^4 = 5.98 \times 10^7\ W/m^2$

The radiated power from 1 cm^2 of sun surface:

$$I \times A = (5.98 \times 10^7)(1 \times 10^{-4}) = 5.98 \times 10^3\ W$$

- For the north-star $I = (5.67 \times 10^{-8})(8300)^4 = 2.69 \times 10^8\ W/m^2$

The radiated power from 1 cm^2 of north-star surface:

$$I \times A = (2.69 \times 10^8)(1 \times 10^{-4}) = 26.9 \times 10^3\ W$$

Example (1.5):

An atomic oscillator emits radiation of 700 nm wavelength how much energy is associated with a photon of light of this wavelength?

Given $h = 6.6261 \times 10^{-34}\ J.s$ and $c = 3 \times 10^8 m/s$.

Solution:

From Planck's the energy of the photon is given by:

$$E = hf = h\frac{c}{\lambda}$$

$$E = \frac{(6.6261 \times 10^{-34})(3 \times 10^8)}{700 \times 10^{-9}} = 2.84 \times 10^{-19}\ J$$

Example (1.6):

What is the maximum kinetic energy of electrons emitted from a zinc surface if they are stopped by a 16 N/C uniform electric field over a distance of 3.0cm? If the electron mass $m_e = 9.1 \times 10^{-31} Kg$ find its velocity.

Solution:

First calculate the voltage $E = V/d$

$$V = Ed = (16\ N/C)(0.03\ m) = 0.48\ V$$

This voltage refer to the stopping voltage V_o

The maximum kinetic energy

$$K_{max} = eV_o = (1.6 \times 10^{-19}\ \mathrm{C})(0.48\ V) = 7.68 \times 10^{-20}\ J$$

Second, calculate the electron velocity using its obtained kinetic energy

$$K_{max} = \frac{1}{2} m_e v^2$$

$$v = \sqrt{\frac{2K_{max}}{m_e}} = \sqrt{\frac{2(7.68 \times 10^{-20})}{9.1 \times 10^{-31}}} = 4.1 \times 10^5 \quad m/s$$

Example (1.7):

What is threshold frequency of a material with a work function of 10eV?

Solution:

1st solution since the value for the work function is given in electron volts, we might as well use the value for Planck's constant that is in $eV.s$.

$$\emptyset = hf_c$$

$$f_c = \emptyset/h = (10\ eV)/(4.14 \times 10^{-15}\ \ eV.s) = 2.4 \times 10^{15}\ \ HZ$$

2nd solution get the a work function in J

$$\emptyset = (10)(1.6 \times 10^{-19}) = 1.6 \times 10^{-18}\ J$$

In this case you could use Planck's constant in J.s

$$f_c = \emptyset/h = (1.6 \times 10^{-18}\ J)/(6.626 \times 10^{-34}\ J.s) = 2.4 \times 10^{15}\ \ HZ$$

Example (1.8):

The work function for a silver surface $\emptyset = 4.73\ \ eV$, find the minimum frequency that light must have to eject electrons from this surface, and then calculate the cutoff wavelength of this light for silver.

Solution:

From the work function we can find the critical frequency

$$f_c = \frac{\emptyset}{h} = \frac{(4.73)(1.6 \times 10^{-19}\ )}{6.626 \times 10^{-34}} = 1.2 \times 10^{15}\ \ HZ$$

The cutoff wavelength could be obtained from the critical frequency

$$\lambda_c = \frac{c}{f_c} = \frac{3 \times 10^8\ m/s}{1.2 \times 10^{15}\ \ HZ} = 2.5 \times 10^{-7}\ \ m = 250\ \ nm$$

Example (1.9):

Calculate the maximum kinetic energy of an electron ejected from a sodium surface whose work function is $\emptyset = 2.28\ eV$, when illuminated by light of wavelength **(a)** 410 nm, **(b)** 550 nm.

Solution:

In first step you must find the energy of incident light corresponding to each wavelength

$$E = hf_c = h\frac{c}{\lambda_c}$$

(a) The energy of ($\lambda = 410\ nm$) will be

$$E = (6.626 \times 10^{-34}\ J.s)\frac{(3 \times 10^8\ m/s)}{(410 \times 10^{-9}\ m)} = 4.85 \times 10^{-19}\ J$$

$$= \frac{(4.85 \times 10^{-19}\ J)}{(1.6 \times 10^{-19}\ c)} = 3.03\ eV$$

This photon energy is greater than the work function of sodium which it can eject electrons from metal surface.

(b) The energy of ($\lambda = 550\ nm$) will be

$$E = (6.626 \times 10^{-34}\ J.s)\frac{\left(3 \times 10^8\ \frac{m}{s}\right)}{(550 \times 10^{-9}\ m)} = 3.61 \times 10^{-19}\ J$$

$$= \frac{(3.61 \times 10^{-19}\ J)}{(1.6 \times 10^{-19}\ c)} = 2.26\ eV$$

This photon energy is less than the work function of sodium (2.28 eV) therefore no electrons are ejected from metal surface.

Example (1.10):

X-ray of 12 pm wavelength are scattered from a target,

(a) Find the wavelength of X-rays scattered through 45° and 90°.

(b) Find the maximum wavelength present in the scattered X-ray.

Solution:

(a) To find the wavelength of X-rays scattered through a given angle:

since $\lambda` - \lambda = \frac{h}{m_o C}(1 - cos\ \theta)$

then $\lambda` = \lambda + \frac{h}{m_o C}(1 - cos\ \theta)$

at $\theta = 45^o$

$$\lambda` = \lambda + \frac{h}{m_o C}(1 - cos\,45) = 12 + 2.243(0.293) = 12.7\,pm$$

at $\theta = 90^o$ $\quad \lambda` = \lambda + \frac{h}{m_o C}(1 - cos\,90) = 12 + 2.243 = 14.243\,pm$

(b) The maximum wavelength present in the scattered X-ray when $\theta = \pi$

then $\Delta\lambda = \lambda` - \lambda = \frac{2h}{m_o C}$

hence $\lambda` = \lambda + \frac{2h}{m_o C} = 12 + 4.86 = 16.86\,pm$

Example (1.11):

An $\boldsymbol{X - ray}$ photon of wavelength $\mathbf{0.01\ nm}$ strikes an electron head on ($\boldsymbol{\theta = 180^o}$) Determine

(a) the change in wavelength of the photon.

(b) the change in energy of the photon.

(c) the kinetic energy imparted to the electron.

Solution:

(a) To find the change in wavelength of the photon $\boldsymbol{\lambda'}$.

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_o c}(1 - \cos\theta)$$

$$= \frac{(6.625 \times 10^{-34}\ \text{J.s})}{(9.11 \times 10^{-31}\ \text{Kg})(3 \times 10^8\ \text{m.s}^{-1})}(1 - \cos 180^o) = 4.848 \times 10^{-12}\ \text{m} = 4848\ \text{nm}$$

then:

$$\lambda' = \lambda + \Delta\lambda = (0.01 \times 10^{-9}\ \text{m}) + (4.848 \times 10^{-12}\ \text{m}) = 104.848 \times 10^{-12}\ \text{m} = 104848\ \text{nm}$$

(b and c) the change in energy of the photon = the kinetic energy imparted to the electron

$$\Delta E = K.E = E - E' = \frac{hc}{\lambda} - \frac{hc}{\lambda'} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right)$$

$$= (1240)\left(\frac{1}{(0.01\ \text{nm})} - \frac{1}{(104848\ \text{nm})}\right) = 123999.99\ \text{eV} = 1.234\ \text{MeV}$$

Problems

1. **If he effective surface temperature of the sun is $5800\,K$, at What wavelength would you expect the sun to radiate most strongly?**

2. At What temperature is cavity radiation most visible to human eye if the eye is most sensitive to yellow - green light, wavelength $= 550\,nm$?

3. Under ideal conditions the normal human eye will record a visual sensation at 550 nm if incident photons are absorbed at a rate as low as $100\ s^{-1}$. To what power level does this correspond

4. An oven with an inside temperature $T_o = 227\,°C$ is in a room having a temperature $T_r = 27\,°C$. There is a - small opening of area $5\,cm^2$ in one side of the oven. How much net power is transferred from the oven to the room? (Hint: Consider both oven and room as cavities)

5. A sodium surface is illuminated with light of wavelength 0.300 μm. The work function for sodium is 2.46 eV.
   a) Calculate the energy of each photon in electron volts,
   b) The maximum kinetic energy of the ejected photoelectrons, and
   c) The cutoff wavelength for sodium.

6. Calculate the energy in electron volts of a photon having a wavelength
   a) in the microwave range, 5.00 cm,
   b) in the visible light range, 500 nm, and
   c) in the x-ray range, 5.00 nm.

7. A certain light source is found to emit radiation whose peak value has a frequency of 1.00×10^{15} Hz. Find the temperature of the source assuming that it is a blackbody radiator.

8. The photocurrent of a photocell is cut off by a retarding potential of 2.92 V for radiation of wavelength 250 nm. Find the work function for the material.

9. The work function for potassium is 2.24 eV. If potassium metal is illuminated with light of wavelength 350 nm, find
   a) The maximum kinetic energy of the photoelectrons and
   b) The cutoff wavelength.

10. **Molybdenum has a work function of 4.2 eV.**
    a) Find the cutoff wavelength and threshold frequency for the photoelectric effect.
    b) Calculate the stopping potential if the incident light has a wavelength of 200 nm.

11. X-rays of wavelength λ_0 =0.200 000 nm are scattered from a block of material. The scattered x-rays are observed at an angle of 45.0° to the incident beam.

- **a)** Calculate the wavelength of the x-rays scattered at this angle.
- **b)** Compute the fractional change in the energy of a photon in the collision.

12. Calculate the fractional change in the wavelength of an Xray of λ equal 0.4 Å that undergoes a 90 ^{0} Compton scattering from an electron.

13. Find the energy and the momentum of X-ray photon whose wavelength is 2 x 10^{-11} m.

14. Abeam of 0.2 MeV photons is scattered by the electrons in a carbon target. Calculate:

- **a) λ associated with these photons**
- **b) λ of those photons that scattered by angle 90°**
- **c) The energy of the scattered photons that emerge at angle of 60° relative to the incident direction.**